$$X(z) = \frac{(1-1/2\,z^{-1})(1-1/4\,z^{-1})(1-1/5\,z)}{(1-1/6\,z)}$$

SEA: $h[m] = \beta^m x[m]$

$$\Rightarrow H(z) = X(z/\beta) = \frac{(1-1/2\,\beta/z)(1-1/4\,\beta/z)(1-1/5\,z/\beta)}{1-1/6\,z/\beta}$$

$$H(z) = \frac{\frac{z-\beta/2}{z}\;\frac{z-\beta/4}{z}\;(1-z/5\beta)}{1-z/6\beta}$$

$$H(z) = \frac{(z-\beta/2)(z-\beta/4)\left(\frac{-1}{5\beta}\right)(z-5\beta)}{\frac{-1}{6\beta}(z-6\beta)\,z^2}$$

$$H(z) = \frac{6}{5}\,\frac{(z-\beta/2)(z-\beta/4)(z-5\beta)}{(z-6\beta)\,z^2}$$

PARA QUE $h[m]$ SEA DE FASE MÍNIMA TODOS SUS CEROS Y POLOS DEBEN ESTAR DENTRO DE LA CIRCUNFERENCIA UNIDAD.

ÚNICO POLO: $\{6\beta\}$ $\rightarrow$ $|6\beta| < 1$

$-1 < 6\beta < 1$

$-1/6 < \beta < 1/6$

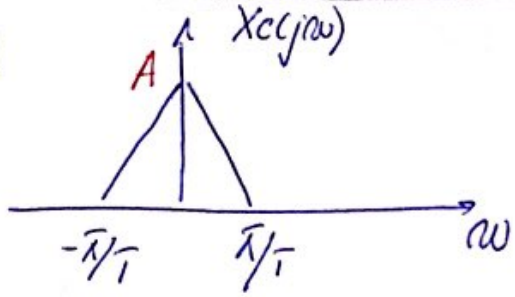
CERO MÁS ALEJADO $\{5\beta\}$ $\Rightarrow$ $|5\beta| < 1$ $\Rightarrow$ $\boxed{\frac{-1}{6} < \beta < \frac{1}{6}}$

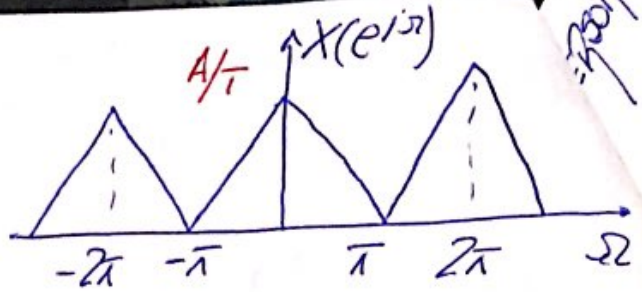
$-1 < 5\beta < 1$

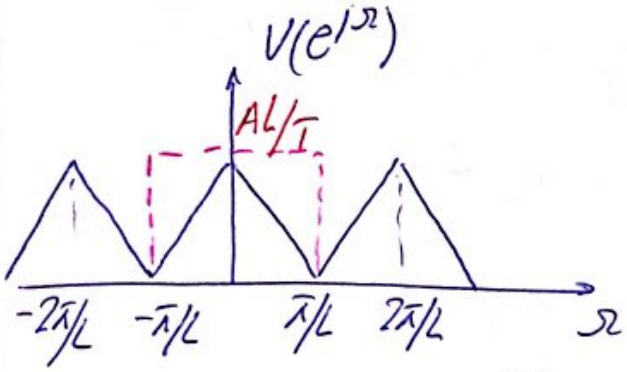
$-1/5 < \beta < 1/5$

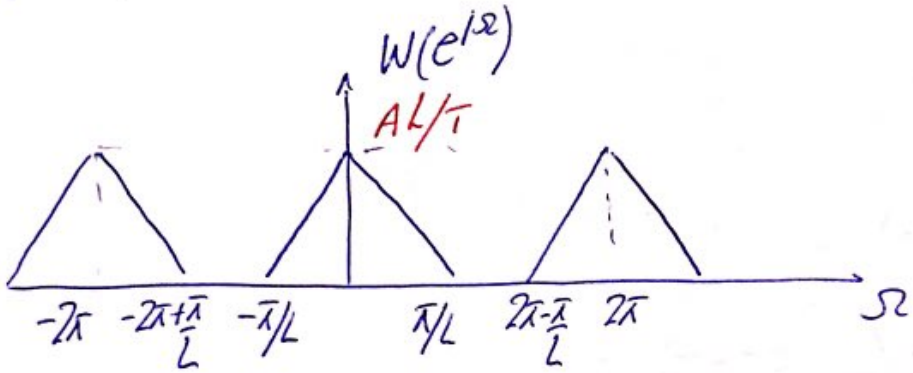
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3)





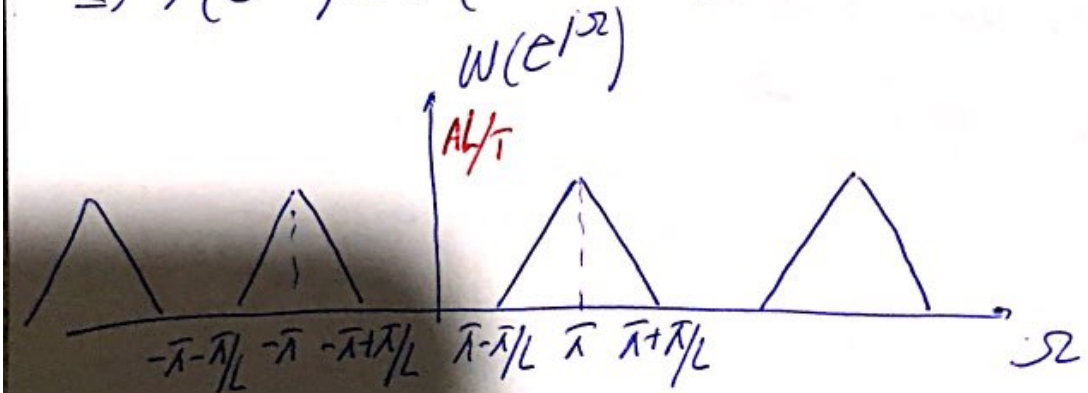




$$(-1)^m = e^{j\pi m}$$

$$\rightarrow g[m] = (-1)^m w[m] = e^{j\pi m} w[m]$$

$$\Rightarrow Y(e^{j\Omega}) = W(e^{j(\Omega-\pi)})$$

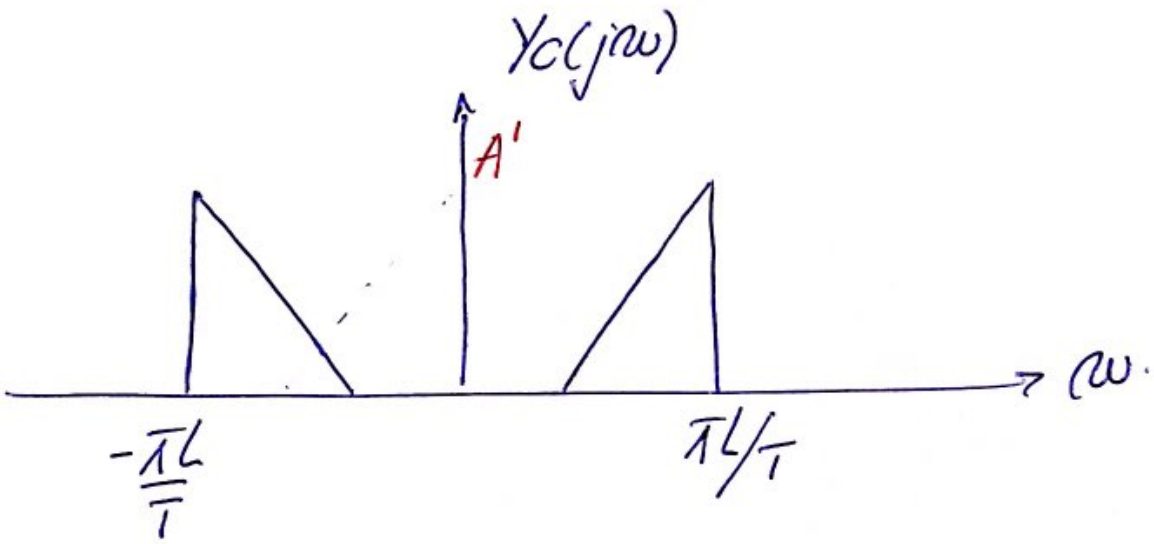


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ERSOR DE TIEMPO DISCRETO A CONTINUO

↳ INTERPOLADOR IDEAL

$$\Rightarrow Hd(e^{j\Omega}) = \begin{cases} T/L ; & |\Omega| < \pi \\ 0 ; & \text{otro caso.} \end{cases}$$



$$A' = \frac{AL}{T}\,\frac{T}{L} = A$$

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